

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Zoology**Paper: ZOB-503 — Biochemistry and Molecular Biology**

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Give brief justification for your answer: [6 × 2 = 12]

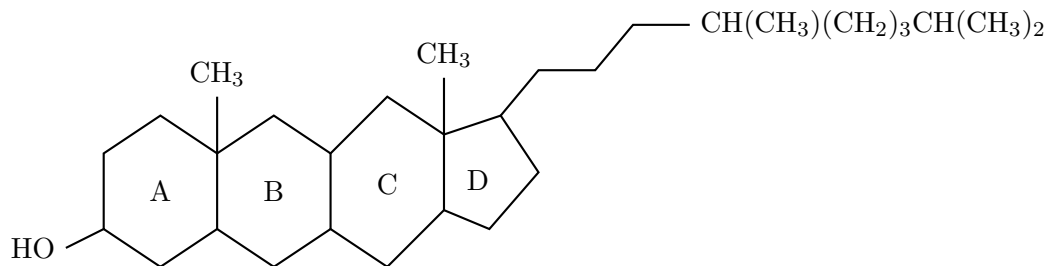
- (i) SD-sequences help in transcription.
- (ii) CPSF facilitates initiation of end replication of DNA.
- (iii) α -helix of a protein is stabilized mainly by intrachain disulphide bonds.
- (iv) Artificially imposed electrochemical gradient can help in ATP synthesis *in vitro* in absence of succinate and oxygen.
- (v) Regulatory enzymes kinetics show a hyperbola curve.
- (vi) pI of an acidic amino acid is determined by $pI = (pK_1 + pK_2 + pK_R)/3$.

- (b) Define the following: [3 × 1 = 3]

- (i) Protein motif
- (ii) Glycosaminoglycans
- (iii) Transition state

- (c) Write the chemical structures of the following: [2 × 2.5 = 5]

- (i) Cholesterol



- (ii) Chitin

Question 2. Discuss the following:

- (a) Write one cycle of sequence of reactions of a peptide with the Edman's reagent. [5]
- (b) Describe the roles of RNA-bound and RNA-free protein complexes in spliceosome mediated splicing. [5]

Question 3. Answer the following:

- (a) Explain the primary level structural constraints of protein folding and the mechanism by which a protein undergoes the folding process. [4]
- (b) With the help of at least two examples, justify the statement, "Quaternary structure provides functional diversity to the protein". [6]

Question 4. Differentiate between the following: [2 × 5 = 10]

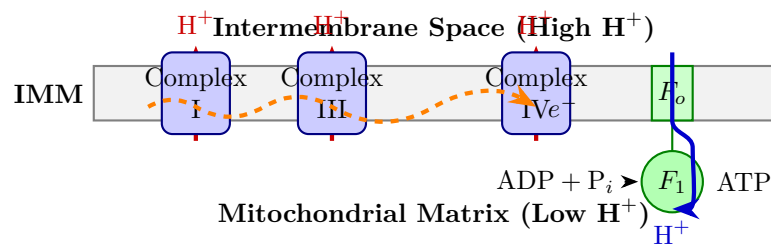
- (a) Uncompetitive and Non-Competitive enzyme inhibition.
- (b) DNA-polymerase- γ and DNA-polymerase- δ .

Question 5. Answer the following:

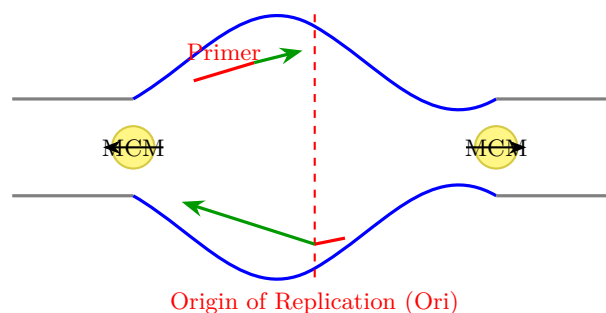
- (a) Illustrate the organization of the pyruvate dehydrogenase complex and its functions. [5]
- (b) Illustrate the process of initiation of translation in eukaryotes. [5]

Question 6. Answer the following:

- (a) Indicating all possible details, illustrate the chemiosmotic theory of oxidative phosphorylation. [5]



- (b) Describe initiation of replication in eukaryotes with a suitable diagram. [5]



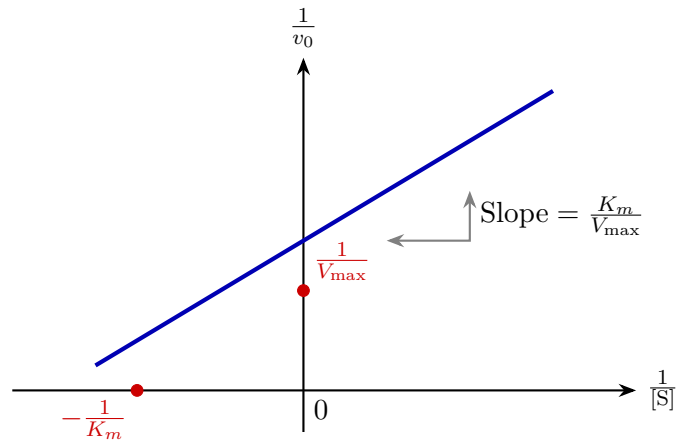
Question 7. Answer the following:

- (a) Write the chemical reactions representing oxidative decarboxylation during Krebs cycle. [5]
- (b) Represent the transcription initiation complex on Promoter Type-II. [5]

Question 8. Explain the following:

[4 + 3 + 3 = 10]

- (a) LB plot helps in better determination of the kinetics parameters of an enzyme.



- (b) Animal cells have stored glycogen rather than starch.
- (c) HDL are known as “good cholesterol”.

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